

A Task-Routed Hybrid Vision–Language System for Medical Visual Question Answering, and a Calibrated Analysis of the Open-Ended Reasoning Ceiling

Saeed Moosivand¹, Fangyijie Wang², and Ziyang Wang¹

¹ Aston University, Birmingham, UK

² University College Dublin, Dublin, Ireland

Abstract. We describe our submission to the MedReason 2026 challenge, a medical visual question answering (VQA) task combining closed-ended multiple-choice (MCQ) and open-ended questions, each requiring an image-grounded reasoning trace. Our final system is a task-routed hybrid built on a single Qwen2.5-VL-7B backbone carrying one Low-Rank Adaptation (LoRA) adapter that is toggled per question type: enabled for MCQ, where fine-tuning raises held-out accuracy from 0.33 to 0.945, and disabled for open-ended questions, where the unmodified base model produces markedly richer image-grounded traces. The routing was motivated by a consistent finding: MCQ-oriented fine-tuning compresses open-ended reasoning traces from a mean of 44.3 to 19.9 words and degrades open-ended quality, a result confirmed by both our held-out validation and the organizers’ own baseline. Beyond the system, we report a set of deliberately rigorous negative results. Self-consistency voting, DINOv3 retrieval augmentation, answer-style prompt engineering, and an open-ended-only LoRA all fail to improve open-ended performance under leakage-controlled evaluation. To evaluate open-ended answers more reliably we build a local large-language-model judge, approximately anchored to the organizers’ single published baseline score; it resolves quality differences that surface-level overlap metrics cannot, and in one case reverses a spurious improvement that a token-overlap metric had reported. Error analysis with this judge points to a likely bottleneck: the 7B backbone confidently names incorrect specific findings on subtle or rare cases, a pattern consistent with a limit in visual-diagnostic perception rather than answer expression or adaptation, though we cannot fully separate perception from medical-knowledge and reasoning factors. All code is released.

Keywords: Medical VQA · Vision–language models · LoRA · LLM-as-judge · Negative results

1 Introduction

Medical visual question answering asks a model to answer clinical questions about a medical image, and increasingly to justify the answer with a reasoning

trace. The MedReason 2026 challenge evaluates this along three equally-weighted axes: MCQ accuracy on closed-ended questions, and two open-ended scores—a ground-truth agreement score (GT) judged by a large language model against a reference answer, and a visual-accuracy score (VA) assessing whether the answer and its reasoning are grounded in the image. Both open-ended scores lie on a 0–4 scale. The final evaluation combines held-out pre-evaluation data with hidden out-of-distribution test data.

Our central finding is that closed-ended and open-ended performance pull the backbone in opposite directions. Parameter-efficient fine-tuning that dramatically improves MCQ accuracy simultaneously degrades open-ended reasoning, because the single-letter MCQ objective rewards terse outputs whose brevity carries over to the open-ended setting. Rather than compromise with one model, we route each question type to the configuration that serves it best, using a single backbone and a toggled adapter to keep the deployed system compact.

We make three contributions. First, a task-routed hybrid VQA system that preserves strong MCQ accuracy while recovering the base model’s superior open-ended behaviour from one shared backbone. Second, a set of leakage-controlled negative results—self-consistency, visual retrieval augmentation, prompt engineering, and open-ended-only fine-tuning—that we believe are individually tempting and collectively informative for future participants. Third, a local LLM judge, approximately anchored to the organizers’ reported baseline, that provides a more discriminative open-ended proxy than surface metrics, together with an error analysis that points to visual-diagnostic perception as a plausible bottleneck.

2 Task and Data

The released training set comprises 17,722 labelled cases: 14,398 closed-ended (MCQ) and 3,324 open-ended. Each case provides one or more images, a question, and—for MCQ—a set of labelled options; open-ended cases provide an expert reference answer and visual description. We hold out 1,000 cases (798 MCQ, 202 open-ended) as a fixed local validation split used for all model selection; no reported model is trained on these cases. All open-ended proxy metrics and judge scores below are computed on the 202 held-out open-ended cases.

3 Method

3.1 Backbone and adaptation

We use Qwen2.5-VL-7B as the backbone. For MCQ we fine-tune a LoRA adapter (rank 16, $\alpha = 32$, dropout 0.05) on the attention projections (q, k, v, o), giving approximately 10M trainable parameters ($\sim 0.12\%$ of the model). Training uses bfloat16, learning rate 5×10^{-5} , effective batch size 16, one epoch, with the loss computed only on target tokens. Open-ended cases are upweighted $3\times$ during this fine-tuning to counter the closed-ended majority.

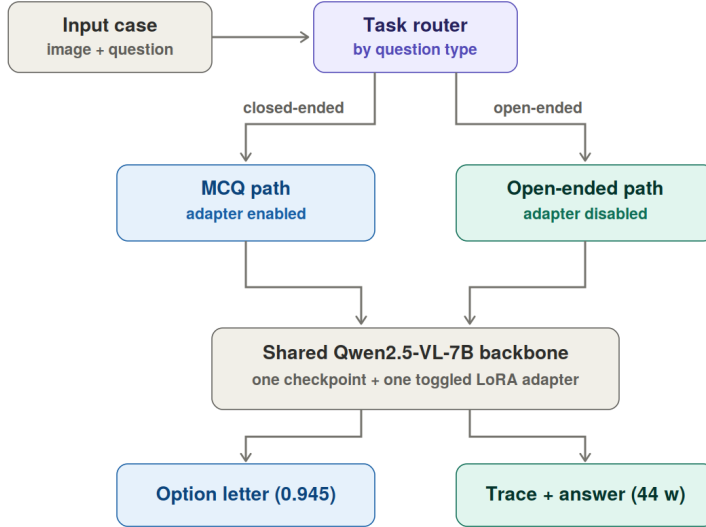


Fig. 1. Task-routed hybrid. A single Qwen2.5-VL-7B backbone carries one LoRA adapter, enabled for MCQ questions and disabled for open-ended questions, producing task-appropriate outputs from one checkpoint.

3.2 Rubric-aware prompting

Prompts are tailored to each task. MCQ prompts request only the option letter; an ablation on the training split showed bare-letter prompting matches structured-JSON prompting in accuracy while running roughly $20\times$ faster, so it is adopted. Open-ended prompts request a compact JSON object with an anatomy-anchored reasoning trace (modality, region, then the specific visible finding) and a single committed answer, instructing the model to describe only what is visible.

3.3 Task-routed hybrid via adapter toggling

The key design decision follows from Section 4: the fine-tuned adapter helps MCQ but harms open-ended answers. We therefore route by task within a single backbone (Fig. 1). The adapter is attached once and enabled for MCQ generation and disabled for open-ended generation, using the adapter enable/disable mechanism of the PEFT library. This yields the fine-tuned model’s MCQ accuracy and the base model’s open-ended behaviour from one checkpoint plus a $\sim 40\text{MB}$ adapter, rather than two full models. All inference is single-pass and deterministic (greedy decoding); the container performs runtime inference with no external network or precomputed outputs.

Table 1. Held-out validation. The hybrid keeps fine-tuned MCQ accuracy and base-model open-ended behaviour. *Base MCQ under our bare-letter prompt; the organizers’ reference setting reports 0.2943.

Configuration	MCQ acc	Open F1	Open cos	Trace words
Base Qwen2.5-VL	0.33*	0.171	0.384	44.3
Fine-tuned (LoRA, v4)	0.945	0.166	0.358	19.9
Hybrid (submitted)	0.945	0.171	0.384	44.3

4 Fine-Tuning Helps MCQ but Harms Open-Ended Reasoning

Fine-tuning raises held-out MCQ accuracy from 0.33 to 0.945. We verify this is not a shortcut: the predicted option-letter distribution over 400 cases (A/B/C/D/E = 72/92/80/83/73) closely matches the gold distribution (67/85/81/88/79), inconsistent with a fixed-letter or positional bias.

On the 202 open-ended cases, however, the fine-tuned model underperforms the off-the-shelf base model on both automatic proxies, and produces far shorter reasoning traces—the mean trace length falls from 44.3 words (base) to 19.9 words (fine-tuned). A natural explanation is the single-token MCQ objective rewarding brevity that transfers to open-ended generation; however, our main adaptation mixes MCQ and (upweighted) open-ended examples, so several factors change together, and the open-ended-only adapter of Section 5.4 also yields terse answers. We therefore state the robust empirical finding—fine-tuning the adapter harms open-ended generation—while noting that adaptation toward short reference targets, not the MCQ objective alone, may drive the effect; isolating the cause is left to future work. This local finding is corroborated by the organizers’ released baseline: the off-the-shelf Qwen2.5-VL scores GT 1.588 and VA 2.696, higher than our fine-tuned single-model submission (GT 1.454, VA 2.296). Both sources agree the base model is the better open-ended generator, which the hybrid exploits.

We validated the toggle end-to-end on a GH200 GPU: with the adapter enabled MCQ accuracy is 0.945 with a balanced letter distribution, and with it disabled the open-ended path reproduces the base model’s 44.3-word traces and 0.171 proxy F1.

5 What Did Not Work: Leakage-Controlled Negative Results

We evaluated four further interventions aimed at open-ended quality. Each is reported because each is a natural thing to try, and each fails under careful evaluation.

Table 2. Retrieval augmentation. The apparent gain is entirely an artifact of self-retrieval.

DINOv3 retrieval	Top-1 sim Open F1	
Full index (leaky)	1.000	0.285
Train-only index (honest)	0.869	0.159
No retrieval (baseline)	—	0.171

5.1 MCQ self-consistency

Sampling five reasoning paths per MCQ (temperature 0.7) and majority-voting the option changed 15 of 400 answers but left accuracy unchanged at 0.945 ($\Delta = +0.000$) for roughly $6\times$ the inference cost. The fine-tuned model is already near-deterministic on MCQ, so voting has no uncertainty to resolve. We therefore did not adopt it.

5.2 DINOv3 retrieval augmentation

We indexed training images with a frozen DINOv3 ViT-L/16 encoder and injected the reference findings of the nearest neighbours as in-context examples. An initial run on the full index appeared to improve open-ended F1 from 0.171 to 0.285, but the mean top-1 retrieval similarity was exactly 1.000—the validation images were retrieving themselves from the index (leakage). Rebuilding the index from the training split only (excluding all validation cases) removed the leakage; mean top-1 similarity fell to 0.869 and open-ended F1 fell to 0.159, below the no-retrieval baseline (Table 2). Visually similar medical images frequently do not share the same specific finding, so retrieved answers act as distractors. We therefore did not adopt retrieval augmentation. This case underlines the necessity of leakage-controlled retrieval evaluation. We note that this variant was assessed primarily with token F1; given the limitations of that metric discussed in Section 6, its rejection should be read together with the leakage evidence rather than from the proxy score alone.

5.3 Answer-style prompt engineering

We compared four open-ended answer styles on the base model: the current prompt, a radiology-impression style, a named-entity style, and a detailed style. The current prompt scored best on token F1 (0.171), with impression (0.099) and named-entity (0.040) worse—the latter two produce very short answers (4.8 and 3.2 words) that overlap metrics penalise, so this comparison is itself limited by the proxy and is reported as indicative rather than conclusive. No prompt variant improved over the existing design, so we retained it.

5.4 Open-ended-only LoRA

To test whether the MCQ majority was responsible for the terseness in Section 4, we trained a LoRA on only the 3,122 open-ended cases, with reference answers as targets. On token F1 this appeared marginally better (0.175 vs 0.171), which would have justified deploying it. As Section 6 shows, the local judge reverses this conclusion: the open-ended-only adapter is in fact substantially worse. Notably, it too produces terse answers, which indicates that adaptation toward short reference targets—rather than the MCQ objective specifically—is a likely driver of the shortening effect. We did not adopt it.

6 A Calibrated Local Judge for Open-Ended Evaluation

Token-overlap F1 and sentence-embedding cosine similarity proved unable to resolve open-ended quality: four very different systems fell within 0.016 F1 of one another, while the challenge’s own GT judge separates teams across a 0.5–2.07 range. We therefore built a local judge: Qwen2.5-32B-Instruct prompted with a 0–4 rubric mirroring the GT definition (clinical agreement with the reference, explicitly independent of length or fluency).

The organizers’ published base-model GT of 1.588 provides a single anchor point. Judging the base model’s answers on our 202 held-out cases yields a mean GT of 1.084. Because this anchor and our score are computed on different splits, it is a weak calibration point rather than a full validation of the judge; the lower absolute value is plausibly explained by our held-out split emphasising harder, subtle-finding questions, but we cannot confirm this from a single cross-split comparison. We therefore treat the judge as an approximately anchored, internally consistent proxy that—unlike the overlap metrics—resolves differences between systems, rather than as a calibrated substitute for the official judge. Stronger validation (human–judge agreement on a subset, scoring with the official judge on the same cases, or agreement with a second independent judge) is left to future work.

Applied to the open-ended-only adapter of Section 5.4, the judge scores it 0.812 versus the base model’s 1.084, and the base model wins 70 of 202 head-to-head cases against 32 for the adapter (Fig. 2, Table 3). This directly reverses the token-F1 conclusion: an approach the overlap metric mildly favoured is in fact clearly worse. The mechanism is visible in answer length—the adapter’s low-scoring answers average 8.5 words against the base model’s 17.7—confirming the terseness hypothesis. This episode is our strongest evidence for the value of a calibrated judge over surface metrics in medical VQA.

7 Error Analysis: A Perceptual, Not Expressive, Ceiling

Stratifying judge scores by question type shows the hardest categories are those demanding identification of a specific finding: diagnosis (mean GT 1.02), location (0.92), and open “other” reasoning questions (0.90) score lowest, while

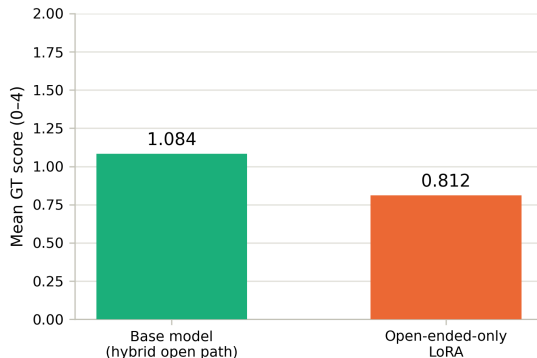


Fig. 2. Calibrated judge GT scores (0–4) on the 202 held-out open-ended cases. The base model, used by the hybrid’s open-ended path, outscores the open-ended-only LoRA, reversing the token-F1 verdict.

Table 3. Calibrated judge on 202 held-out open-ended cases (100 cases tied). The judge reverses the token-F1 verdict of Section 5.4.

System	Mean GT (0–4)	Head-to-head wins
Base model (hybrid open path)	1.084	70 / 202
Open-ended-only LoRA	0.812	32 / 202

broader description (1.21) scores higher. Reading the lowest-scoring cases reveals a consistent pattern: the model does not hedge vaguely—it confidently names an incorrect specific finding. Where the reference is “the right coronary cusp” the model answers “myocardial infarction”; where the reference is “prominent eosinophils” it answers “atypical lymphocytes”; where a finding decreases the model states it increases. These read less as failures of phrasing than as failures to identify subtle or rare findings, a pattern consistent with a limit in the 7B general-purpose backbone’s visual-diagnostic perception. We caution, however, that such errors can also arise from gaps in medical knowledge, question interpretation, or reasoning, and our current analysis does not include the controls (e.g. image-removal, image-permutation, question-only, or stronger-backbone comparisons) that would separate these explanations. We therefore present perception as a plausible dominant bottleneck rather than a definitively localized one.

This offers a plausible account of why every open-ended intervention we tried moved the metric only marginally: prompting, retrieval, and adaptation all act on how the model expresses an answer, whereas the apparent binding constraint is whether it identifies the finding at all. If this account holds, progress on open-ended medical VQA of this style will likely require a stronger or domain-pretrained visual backbone rather than further output-side tuning; confirming it would need the controlled experiments noted above.

8 Conclusion

We presented a task-routed hybrid for medical VQA that keeps a fine-tuned adapter’s MCQ accuracy (0.945) while recovering the base model’s stronger open-ended reasoning from a single backbone. Our broader contribution is the transparent reporting of negative results: four natural interventions fail under leakage-controlled evaluation, and a local LLM judge—approximately anchored to the organizers’ reported baseline—exposes a spurious token-overlap improvement and points to the backbone’s visual-diagnostic perception as a plausible bottleneck. We are careful to distinguish our robust empirical findings from the interpretations that remain to be confirmed with further controls and stronger judge validation. We hope the negative results and the judge methodology are useful to other participants and to work on trustworthy evaluation of open-ended medical VQA. Code is released.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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